

NIDS Session Report

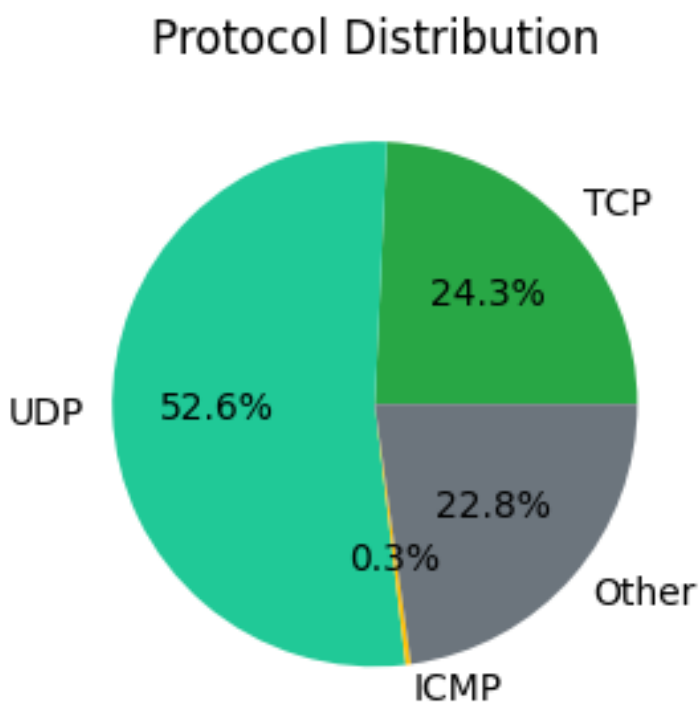
AI-Based Network Intrusion Detection System

This report summarizes your network activity and any suspicious events detected. It's designed to be easy to understand, even if you're not a network expert.

How to Read This Report

- Packets: Each piece of data sent over your network.
- Anomaly: Something unusual or unexpected in the network traffic.
- Alert: A warning about a specific suspicious activity.

Protocol Distribution



Session Summary

Start Time:	1750245768.6763935
Stop Time:	2025-06-18T16:53:52.330712
Uptime:	63 seconds
Packets Processed:	882
Anomalies Detected:	12
Packet Rate:	13.86 packets/sec

Protocol Statistics

TCP PACKETS:	214
UDP PACKETS:	464
ICMP PACKETS:	3
OTHER PACKETS:	201

Security Alerts (4)

Type: Port Scan | Time: 2025-06-18T16:52:50.140730

Details: {'src_ip': '10.0.2.15', 'dst_ip': '8.8.8.8', 'dst_port': 443, 'protocol': 17}

Explanation: A device tried to quickly check many different ports on another device. Hackers often do this to find weak spots before launching an attack.

Type: Ddos Indicator | Time: 2025-06-18T16:52:50.256862

Details: {'src_ip': None, 'dst_ip': None, 'packet_size': 1514, 'protocol': None}

Explanation: A large number of packets were sent to a single device, possibly overwhelming it. This is a common sign of a Distributed Denial of Service (DDoS) attack, which can take down services.

Type: MI Anomaly | Time: 2025-06-18T16:52:53.719715

Details: {'anomaly_score': -7.301462642705259, 'features': {'packet_count': 100, 'total_bytes': 40699, 'avg_packet_length': np.float64(406.99), 'std_packet_length': np.float64(505.64688261671307), 'min_packet_length': 60, 'max_packet_length': 1514, 'avg_payload_size': np.float64(221.26), 'std_payload_size': np.float64(421.13450630410233), 'total_payload_size': 22126, 'tcp_count': 2, 'udp_count': 58, 'icmp_count': 1, 'other_protocol_count': 0, 'protocol_diversity': 3, 'tcp_ratio': 0.03278688524590164, 'udp_ratio': 0.9508196721311475, 'icmp_ratio': 0.01639344262295082, 'other_protocol_ratio': 0.0, 'unique_src_ports': 8, 'unique_dst_ports': 10, 'total_ports_accessed': 10, 'common_port_ratio': 0.4166666666666667, 'high_port_ratio': 0.5833333333333334, 'avg_dst_port': np.float64(31815.366666666665), 'std_dst_port': np.float64(27376.408695545797), 'min_dst_port': 53, 'max_dst_port': 59191, 'unique_connections': 9, 'total_connection_attempts': 61, 'avg_packets_per_connection': 6.777777777777778, 'connection_diversity': 0.09, 'max_packets_per_connection': 18, 'unique_src_ips': 6, 'unique_dst_ips': 5, 'src_ip_diversity': 0.06, 'dst_ip_diversity': 0.05, 'private_src_ip_ratio': 0.5, 'private_dst_ip_ratio': 0.4, 'packet_rate': 60.89674336380321, 'inter_arrival_time_mean': np.float64(0.016587110480876886), 'inter_arrival_time_std': np.float64(0.10882610761628672), 'burst_count': 77, 'burst_ratio': 0.7777777777777778, 'payload_size_mean': np.float64(221.26), 'payload_size_std': np.float64(421.13450630410233), 'zero_payload_ratio': 0.39, 'large_payload_ratio': 0.12, 'payload_size_entropy': np.float64(1.2510469589756323), 'syn_count': 2, 'fin_count': 0, 'rst_count': 0, 'psh_count': 0, 'ack_count': 2, 'urg_count': 0, 'syn_ratio': 1.0, 'fin_ratio': 0.0, 'rst_ratio': 0.0, 'psh_ratio': 0.0, 'ack_ratio': 1.0, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:52:53.719700', 'packet_count': 100, 'protocol_distribution': {'tcp': 2, 'udp': 58, 'icmp': 1}, 'connection_info': {'unique_connections': 9, 'unique_src_ips': 6, 'unique_dst_ips': 5}}

Explanation: The AI detected network activity that looks very different from normal. This could mean someone is trying something unusual or potentially harmful on your network.

Type: Port Scan | Time: 2025-06-18T16:53:50.263857

Details: {'src_ip': '10.0.2.15', 'dst_ip': '142.250.182.67', 'dst_port': 443, 'protocol': 17}

Explanation: A device tried to quickly check many different ports on another device. Hackers often do this to find weak spots before launching an attack.

What Should I Do?

- If you see many alerts: Consider checking your devices for unusual activity or malware.
- If you see DDoS or Port Scan alerts: Make sure your firewall is active and up-to-date.
- If unsure: Share this report with your IT support or a cybersecurity professional.

Glossary of Terms

Packets Processed:	Total number of network packets analyzed by the NIDS during this session.
Anomalies Detected:	Number of network events flagged as suspicious or abnormal by the NIDS.
Packet Rate:	Current rate at which packets are being processed.
Uptime:	How long the NIDS was running.
TCP/UDP/ICMP/Other:	Number of packets for each protocol.
Port Scan:	A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.
DDoS Indicator:	Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.
Unusual Protocol:	Unusual network protocol detected. This could indicate suspicious or unauthorized network activity.
ML Anomaly:	The machine learning model detected traffic that is statistically unusual compared to normal patterns.